Jianxin Zhao

ML Systems Engineer · LLM/VLM Inference · ML Frameworks

PhD in Computer Science, University of Cambridge | Numerical computing and ML systems since 2016
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Highlights

- **Inference systems:** lead the OverDrive project on real-time, on-board inference for VLM/VLA-based autonomous-driving models; profiled 4 open-source driving models on GPU and reduced the combined runtime of preprocessing and vision encoding by 38.5% in a stage-level benchmark.
- **ML frameworks:** core developer and project lead of Owl, an open-source numerical and deep-learning library for OCaml. Co-designed Owl's computation-graph abstraction and led ONNX export and memory optimisation; graph-level memory reuse reduced Mask R-CNN CPU inference memory by 86.6%.
- **Research and delivery:** 25+ publications (incl. IEEE TMC, TSC, TKDE, ICDE) and 4 books; delivered a real-time water-level prediction module for Rhine ports and a legal knowledge-base system piloted at a law firm.

Experience and Selected Projects

Karlsruhe Institute of Technology (KIT), Germany Oct 2023 – Oct 2026
Postdoctoral Researcher · AI Systems for Autonomous Driving

- **OverDrive (lead):** inference scheduling for VLM/VLA driving models under scenario-dependent latency SLOs, so that planning results arrive in time. Workloads: 4 open-source driving models built on Qwen2.5-VL, InternVL2, LLaVA and others.
  - Led the development of a cross-model execution measurement platform: 185K inference samples and 8,652 CARLA reference-controller simulations.
  - Profiled GPU timelines and kernels with Nsight Systems and Nsight Compute to locate KV-cache, memory-bandwidth and compute bottlenecks, and evaluated cross-frame and operator-level concurrency on GPU.
  - Overlapped next-frame CPU preprocessing with current-frame vision encoding, reducing paired-stage runtime by 38.5% in an isolated benchmark on an RTX 4090. GPU operator-level pipelining and end-to-end simulator evaluation are in progress.
- **Industry collaboration (Covadonga, Germany):** developed the real-time water-level prediction module of a monitoring system for several Rhine inland ports, now in live operation.
- First author of *Autonomous Driving: Technology, Business, and Regulation* (Wiley, October 2026). Contributed to the C2CBridge 2 project (German Federal Ministry for Transport); co-teach two Master's courses.

Owl: numerical computing and deep learning for OCaml Core developer · Project lead 2017 – present
Open-source library for numerical computing, automatic differentiation and neural networks in OCaml (about 236K lines of code; 1.3k+ GitHub stars), running on x86/ARM CPUs and GPUs and used by companies including in quantitative finance. A long-running open-source project alongside my PhD and postdoctoral roles.

- Co-designed the computation-graph abstraction, which separates definition from execution to enable lazy evaluation, graph optimisation and compilation.
- Led graph-level memory reuse on top of it: on CPU, compared with execution without graph optimisation, Mask R-CNN inference memory fell by 86.6% (6.5 GB to 0.87 GB) and time by 24.2% including graph construction.
- Led owl_symbolic, which exports Owl computation graphs to ONNX and validates the exported models with ONNX Runtime, and an Owl-to-TensorFlow computation-graph converter.
- Implemented the convolution operators in C with automatic parameter tuning; on ARM, convolution matched the Eigen implementation that TensorFlow used on CPU at the time.
- Led the Zoo deployment framework with JavaScript and MirageOS unikernel backends; explored GPU/TPU execution (OCaml Workshop @ ICFP 2019 talk).
- Project lead and release manager since 2020 (releases 1.0–1.2, code review, CI/CD and community); secured two OCaml Software Foundation grants; led the official tutorial and co-authored the Apress and Springer books on Owl.

Beijing Institute of Technology (BIT), China Feb 2021 – Apr 2023
Postdoctoral Researcher · Edge Intelligence and Industrial AI

- **Industrial IoT data lake (National Key R&D Program sub-project, core member):** researched transfer-learning anomaly detection for label-scarce target domains such as industrial inspection, using cross-domain representation alignment (IEEE TKDE 2023, corresponding author).
- **Compilation and operator optimisation:** TVM-based cross-platform deep-learning compilation with cost-model pre-training; automatic operator performance tuning on edge devices.
- **Edge-intelligence research (collaboration ongoing):** synchronisation and resource optimisation for distributed and federated learning, and multi-agent reinforcement learning for crowdsensing. First or corresponding author of papers in IEEE TMC, TSC, TITS, JSTSP, ICDE and others; 4 Chinese invention patents (first inventor).

KnowledgeBase: legal knowledge-base and agent system Independent project · Law-firm pilot 2026 – present

- Built and deployed an AWS-hosted system piloted at a leading Chinese law firm, integrating hybrid retrieval, a multi-level knowledge graph, MCP tools and a LibreChat front end (with its LangGraph-based agent framework); iterating on lawyers' feedback.
- **Retrieval design evaluation:** probability simulations at fixed corpus size across five relevance densities showed a **32–77% lower simulated query token budget** than flat retrieval at the same 90% mean recall.
- **Cost and processing efficiency:** optimised server configuration to reduce server costs by approximately **67%**; replaced full recomputation with incremental entity updates, reducing cumulative source-material processing from $O(n^2)$ **to** $O(n)$ as sources are appended; introduced asynchronous Batch API processing for non-real-time workloads, with **50% lower input/output token unit prices** for eligible background calls.

FastAPI · PostgreSQL + pgvector · Docker Compose · MCP · LibreChat · AWS

Education

PhD in Computer Science, University of Cambridge Oct 2015 – Jan 2021
Supervisor: Prof. Jon Crowcroft FRS FREng; China Scholarship Council full scholarship

MEng in Software Engineering, Beijing Institute of Technology Sep 2013 – Jul 2015

BSc in Software Engineering, Beijing Institute of Technology Sep 2009 – Jul 2013

Skills

- **Inference optimisation:** PyTorch, Nsight Systems / Nsight Compute, CPU–GPU pipelining, GPU concurrency analysis, output-consistency testing.
- **ML frameworks:** C numerical kernels, computation graphs, ONNX model export, memory optimisation.
- **Working knowledge:** vLLM and SGLang architecture (PagedAttention, RadixAttention), prefill/decode scheduling, speculative decoding, quantisation.
- **AI-assisted development:** multi-agent development workflow with role separation, milestone gates and evidence logs, using Claude Code and ChatGPT.
- **Languages and tools:** Python, C, OCaml, MATLAB; Docker, AWS.
- **Certifications: AWS Certified Solutions Architect – Professional**; AWS Certified AI Practitioner; NVIDIA-Certified Associate: Generative AI LLMs.
- **Languages:** English (C2), Chinese (native), Japanese (B1), German (A1).

Selected Publications and Books

- 25+ publications; own name in bold, * corresponding author. Selected:
  - **J. Zhao**, P. Vandenhove, P. Xu, et al. Parallel and Memory-Efficient Distributed Edge Learning in B5G IoT Networks. **IEEE JSTSP 2023**
  - **J. Zhao**, R. Han, Y. Yang, et al. Federated Learning With Heterogeneity-Aware Probabilistic Synchronous Parallel on Edge. **IEEE TSC 2022**
  - X. Wu, **J. Zhao**, P. Xu, C. H. Liu. Cross-Platform Deep Learning Compilation with Model-Specific Cost Model Pre-training. **ICMLC 2025**
  - Q. Zhao, C. H. Liu, **J. Zhao***, et al. UAV Carrier Enabled Vehicular Crowdsensing by Multi-Agent Reinforcement Learning With Mutual Policy Divergence and Attentive Memory Update. **IEEE TMC 2026**
- Four books: first author of *Autonomous Driving: Technology, Business, and Regulation* (Wiley, 2026); co-author of two books on Owl (Apress 2022, Springer 2022) and one on enterprise analytics (Springer 2024).
- PI of an NSFC Young Scientists Fund project; recipient of the China Postdoctoral International Exchange Program fellowship; 4 Chinese invention patents (first inventor); iENA Nuremberg Gold Medal 2025 (team award).

Appendix: Grants, Books, Publications and Teaching

Grants and projects

amounts are total project budgets

2025.01–2026.12 Industry collaboration with Covadonga (Germany): water-level prediction for a Rhine inland-port monitoring system

2024.08–2026.07 C2CBridge 2, German Federal Ministry for Transport (BMV): autonomous electric mobility linking rural and urban areas (€12.7M, participant)

2024.01–2026.12 NSFC Young Scientists Fund: Edge-side deep model training for mobile crowdsensing (RMB 300K, PI)

2022.01–2026.12 NSFC Key Program: Fundamental theories and technologies for edge-side big-data computing (RMB 2.98M, core member)

2022.01–2025.12 NSFC Joint Fund Key Program: Key technologies for edge crowd-intelligence sensing and capability enhancement (RMB 2.6M, core member)

2021.12–2024.11 National Key R&D Program (Industrial Software) sub-project: Big-data lake system for heterogeneous industrial IoT (RMB 3.518M, core member)

2021.01–2023.01 China Postdoctoral International Exchange Program fellowship (RMB 400K, recipient)

2016.10–2017.10 EPSRC Databox: privacy-aware infrastructure for managing personal data (£1.2385M, core member)

Books

- **J. Zhao**, L. Wang, A. Vinel. *Autonomous Driving: Technology, Business, and Regulation*. Wiley, 2026. Foreword by Dr. Johann Prenninger, Head of Analytics / Machine Learning, BMW AG, who calls it "an essential guide to entering the AD field".
- L. Wang, **J. Zhao**. *Strategic Blueprint for Enterprise Analytics*. Springer, 2024.
- L. Wang, **J. Zhao**. *Architecture of Advanced Numerical Analysis Systems: Designing a Scientific Computing System Using OCaml*. Apress, 2022 (open access).
- L. Wang, **J. Zhao**, R. Mortier. *OCaml Scientific Computing: Functional Programming in Data Science and Artificial Intelligence*. Springer, 2022; second edition invited.

Publications

own name in bold; * corresponding author

Journal articles

- F. Zhou, C. H. Liu, **J. Zhao***, et al. AoI and Latency-Aware Air-Ground Vehicular Crowdsensing by Sequential Multi-Agent Deep Reinforcement Learning. **IEEE TMC 2026**
- Q. Zhao, C. H. Liu, **J. Zhao***, et al. UAV Carrier Enabled Vehicular Crowdsensing by Multi-Agent Reinforcement Learning With Mutual Policy Divergence and Attentive Memory Update. **IEEE TMC 2026**
- H. Yang, C. H. Liu, G. Li, H. Wang, **J. Zhao**, et al. Indoor Fingerprint Collection Under Environment Changes by Vehicular Crowdsensing: A Bayesian Reinforcement Learning Approach. **IEEE TMC 2026**
- L. Wan, **J. Zhao**, et al. A Systematic Literature Review on Vehicular Collaborative Perception — A Computer Vision Perspective. **IEEE TITS 2026**
- X. Wu, S. Du, K. Qiu, S. Wen, H. Hou, Y. Liu, **J. Zhao**, et al. TrustFCL: Task-grained Model Assessment and Aggregation for Secure Federated Continual Learning. **IJOCTA 2026**
- S. Wang*, **J. Zhao**, C. Liu, W. Wu, Z. Liu. Binary Code Fuzzing Based on Deep Reinforcement Learning (in Chinese). **Computer Science 2024**
- S. Li, S. Li, M. Xie, K. Gong, **J. Zhao***, et al. End-to-End Transferable Anomaly Detection via Multi-Spectral Cross-Domain Representation Alignment. **IEEE TKDE 2023**
- **J. Zhao**, P. Vandenhove, P. Xu, et al. Parallel and Memory-Efficient Distributed Edge Learning in B5G IoT Networks. **IEEE JSTSP 2023**
- **J. Zhao**, Y. Feng, X. Chang, et al. Energy-Efficient and Fair IoT Data Distribution in Decentralised Federated Learning. **IEEE TNSE 2023**
- **J. Zhao**, X. Chang, Y. Feng, C. H. Liu, N. Liu. Participant Selection for Federated Learning With Heterogeneous Data in Intelligent Transport System. **IEEE TITS 2023**
- **J. Zhao**, R. Han, Y. Yang, et al. Federated Learning With Heterogeneity-Aware Probabilistic Synchronous Parallel on Edge. **IEEE TSC 2022**
- **J. Zhao**, Y. Feng, X. Chang, C. H. Liu. Energy-efficient client selection in federated learning with heterogeneous data on edge. **Peer-to-Peer Networking and Applications 2022**
- H. Gao, C. H. Liu, W. Wang, **J. Zhao**, et al. A Survey of Incentive Mechanisms for Participatory Sensing. **IEEE Communications Surveys & Tutorials 2015**

Conference and workshop papers

- Z. Xia, **J. Zhao**, A. Vinel. Dynamic-PBFT: Enhanced Consensus in Decentralized Federated Averaging for V2V Networks. **IEEE VTC 2025-Fall**
- X. Wu, **J. Zhao**, P. Xu, C. H. Liu. Cross-Platform Deep Learning Compilation with Model-Specific Cost Model Pre-training. **ICMLC 2025**
- **J. Zhao**, M. Lin, A. Vinel. V2X-Based Decentralized Singular Value Decomposition in Dynamic Vehicular Environment. **ECCV 2024 Workshops**
- Y. Ye, C. H. Liu, Z. Dai, **J. Zhao***, et al. Exploring both Individuality and Cooperation for Air-Ground Spatial Crowdsourcing by Multi-Agent Deep Reinforcement Learning. **ICDE 2023**
- Y. Wang, J. Wu, X. Hua, C. H. Liu, G. Li, **J. Zhao**, et al. Air-Ground Spatial Crowdsourcing with UAV Carriers by Geometric Graph Convolutional Multi-Agent Deep Reinforcement Learning. **ICDE 2023**
- P. Xu, X. Chang, **J. Zhao**, C. H. Liu. Automatic Operator Performance Tuning in a Machine Learning System on Edge. **ICPADS 2022**
- **J. Zhao**, R. Mortier, J. Crowcroft, L. Wang. Privacy-Preserving Machine Learning Based Data Analytics on Edge Devices. **AAAI/ACM AIES 2018**

In preparation and talks

- **J. Zhao**, et al. OverDrive: Deadline-Aware Lifecycle Management of VLM-based Driving Model Plans. (in preparation)
- **J. Zhao**. Executing Owl Computation on GPU and TPU. **OCaml Workshop @ ICFP 2019** (talk)

Teaching and supervision

- Co-taught at KIT with Prof. Alexey Vinel and colleagues: *Collective Perception in Autonomous Driving* (Master's lecture and exercises, summer 2024 and 2025); *Artificial Intelligence for Autonomous Driving* (Master's seminar, winter 2024/25 and 2025/26).
- Co-supervised 7 Master's theses at KIT (2), BIT (3) and Cambridge (2), on pedestrian intention prediction, V2X federated learning, federated learning and deep-learning compilation; supervised undergraduates in Computer Networking at Cambridge (2016).

Patents and awards

- **Chinese invention patents (4, all as first inventor):** client selection for edge-side federated learning under heterogeneous data (ZL202111498897.1); large-scale edge machine-learning training based on probabilistic sampling (ZL202110285186.X); game-theoretic distributed learning for crowdsourced image recognition on mobile devices (ZL202210945014.5); distributed learning for edge-side image recognition in intelligent transportation (ZL202211441906.8).
- **Software copyrights (6, China):** implementations of 2021–2023 distributed and federated learning research.
- **Awards:** Gold Medal (team award, as a member of the inventor team), iENA International Exhibition of Inventions, Nuremberg 2025 (air–ground collaborative mobile crowdsensing system and method for smart cities); China Scholarship Council full PhD scholarship.